

Lab 3

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Submission Instructions

This lab (and all labs) will be graded for correctness. To submit the lab:

1. Once all code is completed and saved, in the RStudio ribbon, click Knit > Knit to PDF.
2. If you have trouble knitting **anything** to PDF, please work with the Instructor or GSI to resolve the issue. It is okay to instead knit to HTML, then convert to PDF using Adobe or other software.
3. If there is any handwritten work for the lab, combine the handwritten work with the knitted PDF into a single PDF file.
4. Submit the PDF to Gradescope, **selecting** pages for questions. *You will lose half a point on the lab if you don't select pages.*

Lab 3 is due to Gradescope by Sunday, February 5th at 11:59PM PT.

This lab will introduce some common R workflows for exploring multivariate relationships and running multiple linear regression.

Read and explore the data

We will explore the impact of several continuous housing covariates on Boston rent (in 1978).

Note: For some interesting Bay Area rental data over time, you can take a look at this UC Berkeley data on gentrification and displacement in the Bay Area. We can access this data via the Tidy Tuesday package, which is one of the provided data resources for your group project. This data does not have many continuous covariates, so we avoid analyzing it for now.

```
tuesdata <- tidyuesdayR::tt_load('2022-07-05')
```

```
## --- Compiling #TidyTuesday Information for 2022-07-05 ----
```

```
## --- There are 3 files available ---
```

```
## --- Starting Download ---
```

```
##
```

```
## Downloading file 1 of 3: 'rent.csv'
```

```
## Downloading file 2 of 3: 'sf_permits.csv'
```

```
## Downloading file 3 of 3: 'new_construction.csv'
```

```
## --- Download complete ---
```

```
## the data are contained as list elements:  
bayarea_rent <- tuesdata$rent  
sf_permits <- tuesdata$sf_permits  
bayarea_construction <- tuesdata$new_construction
```

Back to the data we will explore in the lab, here's how to load it in:

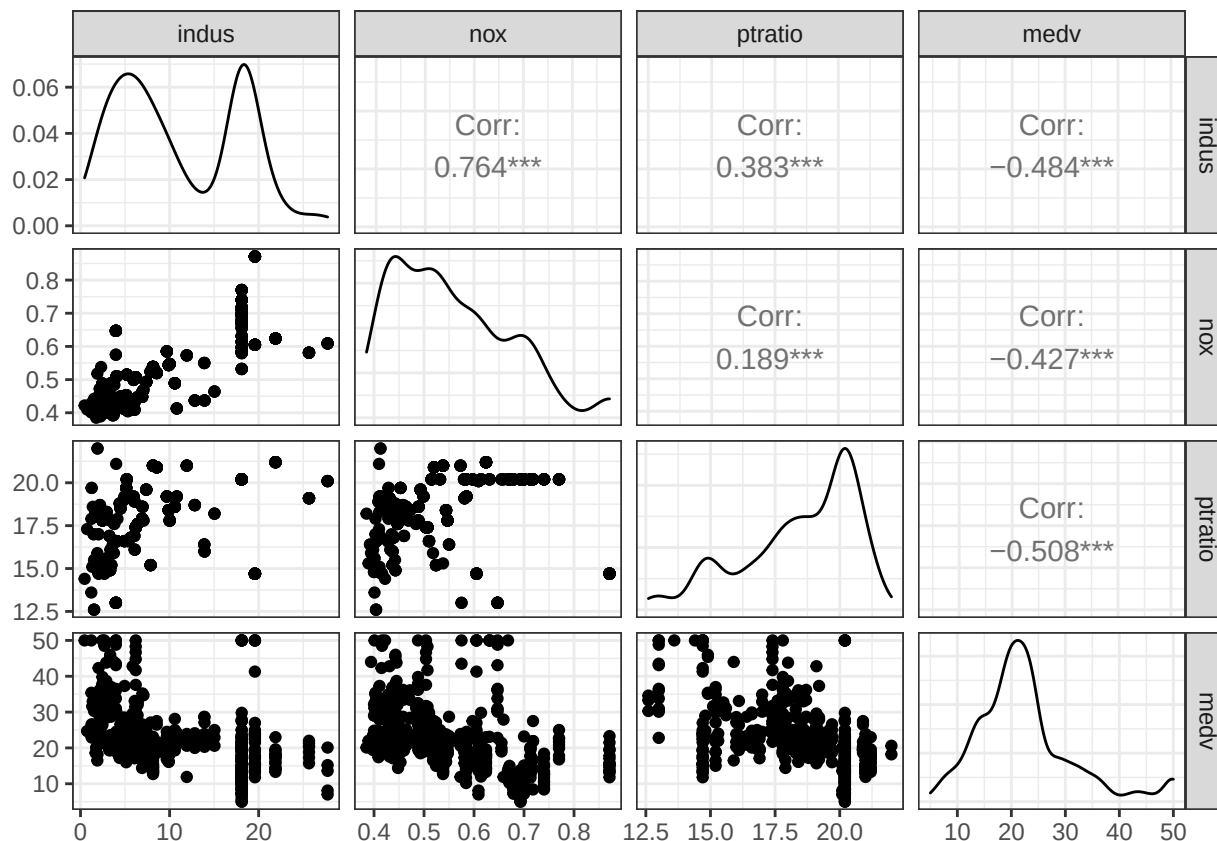
```
boston_housing <- read.table("BostonHousing.txt", header = TRUE, sep = ",")
```

The data dictionary is as follows:

- CRIM - per capita crime rate by town
- ZN - proportion of residential land zoned for lots over 25,000 sq.ft.
- INDUS - proportion of non-retail business acres per town.
- CHAS - Charles River dummy variable (1 if tract bounds river; 0 otherwise)
- NOX - nitric oxides concentration (parts per 10 million)
- RM - average number of rooms per dwelling
- AGE - proportion of owner-occupied units built prior to 1940
- DIS - weighted distances to five Boston employment centres
- RAD - index of accessibility to radial highways
- TAX - full-value property-tax rate per \$10,000
- PTRATIO - pupil-teacher ratio by town
- PWORK - percentage of population working class
- MEDV - Median value of homes (in \$1000's)

Q1: Use the `ggpairs()` function to examine the relationships between INDUS, NOX, PTRATIO, and MEDV. Any trends or nonlinearity you see?

```
### YOUR CODE HERE ###  
ggpairs(select(boston_housing, indus, nox, ptratio, medv)) + theme_bw()
```



ANSWER: Indus univariate distribution is bimodal. Medv scatterplots are non-linear. Ptratio has a non linear association with nox and indus.

Note: Ideally, to compare relationships across scatter plots, we should normalize the variables, i.e., subtract the mean and divide by the SD. This standardization does not impact the correlation. It puts all the measurements in standard units so scatter plots are more easily compared. The automatic scaling of the pairs of scatter plots in the `ggpairs()` output effectively do this for us.

Q2: Which range of Box-Cox transformations might you suggest to someone looking to linearize the trend between nox and ptratio?

ANSWER: ptratio: $p > 1$, nox: $p < 1$

The top right outputs of `ggpairs()` gives us the bivariate correlations, . Correlation measures linear association, but it does not determine whether the data are linearly associated. For example, all of the plotted pairs have very significant Pearson correlation (triple asterisk $\rightarrow p \leq 0.001$), but we can clearly visually detect nonlinearities.

We will ignore transformations for the remainder of the lab, although this would certainly be something to keep considering in practice.

Two Variable Regression

Consider how well the pair of variables proportion non-retail and pollution might explain the variation median home value.

Q3: Run a linear regression of INDUS and NOX on median home value, and print a summary of the model.

```
## YOUR CODE HERE ###
model = lm(medv ~ indus + nox, data = boston_housing)
summary(model)

##
## Call:
## lm(formula = medv ~ indus + nox, data = boston_housing)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -12.48  -4.87  -1.61   3.02  32.30
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  34.2883     2.0762   16.51 < 2e-16 ***
## indus        -0.5062     0.0806   -6.28 7.3e-10 ***
## nox         -11.0291     4.7721   -2.31  0.021 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 8.02 on 503 degrees of freedom
## Multiple R-squared:  0.242, Adjusted R-squared:  0.239
## F-statistic: 80.3 on 2 and 503 DF, p-value: <2e-16
```

How do we interpret these coefficients?

The coefficient for non-retail proportion is interpreted in the presence of pollution. For example, if we choose a value for pollution of 0.5 and plug this value into the fitted model, then the relationship between INDUS and MEDV is:

$$34.2883 + 0.5 * (-11.0291) + indus * (-0.5062)$$

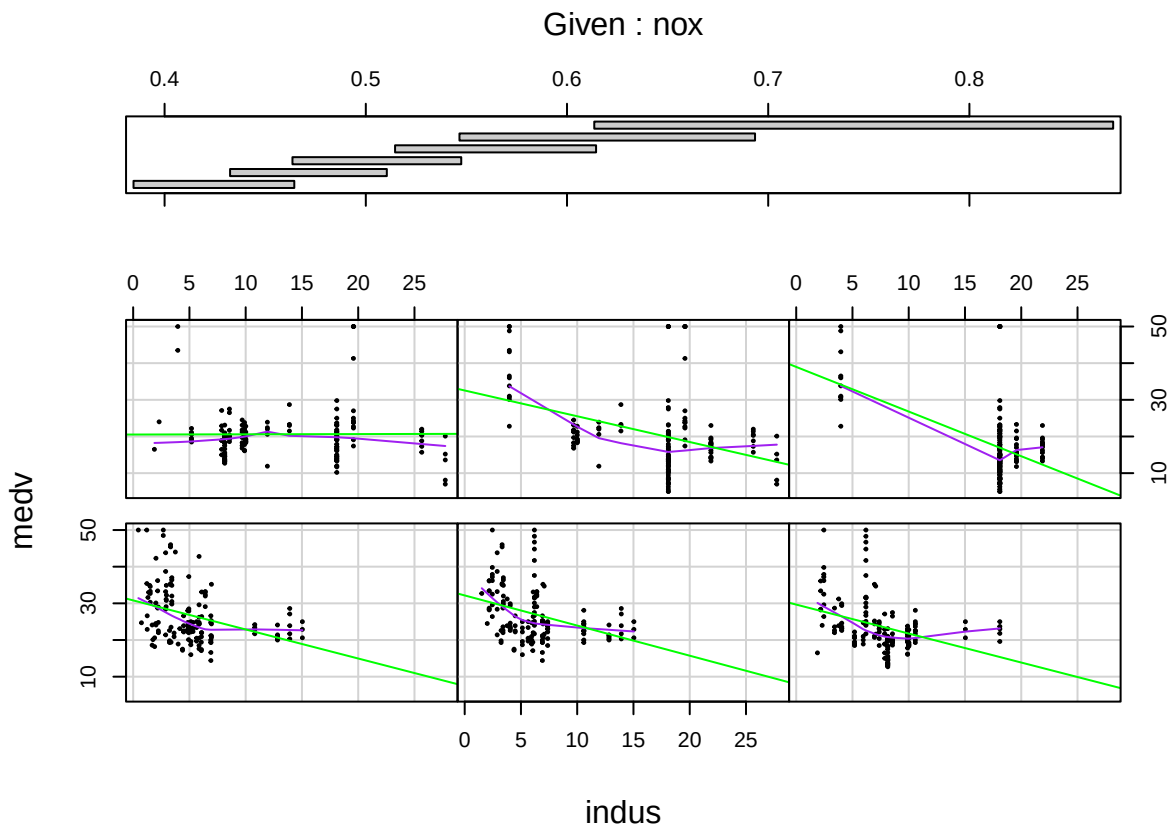
For pollution of 0.4, the relationship is:

$$34.2883 + 0.4 * (-11.0291) + indus * (-0.5062)$$

Notice that the intercept changes, but the slope of these two lines remains the same. The implication is that the linear relationship remains the same for all values of NOX, i.e., the slope does not change, regardless of pollution levels.

We can examine this property with a coplot that conditions on various ranges of NOX. In short, the data is split-up into small ranges of NOX, and a corresponding scatter plot and regression line of INDUS on MEDV is visualized for each range.

```
coplot(medv ~ indus | nox, data = boston_housing,
       panel=function(x,y,...) {
         panel.smooth(x,y,span=.8,iter=5, col.smooth = "purple", ...)
         abline(lm(y ~ x), col="green")
       },
       pch = 19, cex = 0.3,
       ylab = "medv")
```



Q4: What are the green lines? Roughly speaking, what are the purple lines? Does the assumption of “same slope” for INDUS conditional on various values of NOX appear to be violated? Why?

ANSWER: Regression line of best fit conditioning on a range of nox values. A nonlinear relationship/cure between indus and medv conditioning on a range of nox values. Yes this assumption is violated as all the different slopes are different.

Three Variable Regression

Q5: Run a linear regression of INDUS, NOX, and PTRATIO on median home value, and print a summary of the model.

```
## YOUR CODE HERE ###
model2 = lm(medv ~ indus + nox + ptratio, data = boston_housing)
summary(model2)
```

```
##
## Call:
## lm(formula = medv ~ indus + nox + ptratio, data = boston_housing)
##
```

```
## Residuals:
##      Min       1Q   Median       3Q      Max
## -14.38  -4.52  -1.10   2.77  33.99
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  67.1837     3.6466   18.42 < 2e-16 ***
## indus        -0.1933     0.0788   -2.45  0.015 *
## nox         -19.0706     4.3901   -4.34 1.7e-05 ***
## ptratio      -1.7295     0.1643  -10.53 < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 7.27 on 502 degrees of freedom
## Multiple R-squared:  0.379, Adjusted R-squared:  0.375
## F-statistic: 102 on 3 and 502 DF,  p-value: <2e-16
```

Q6: Compare the significance of the coefficients on INDUS and NOX between this regression and the 2-variate regression in Q3. What implications might this have for a causal study of the effect of non-retail industry (INDUS) on home value in Boston?

ANSWER: Significance went down for indus and went up for nox. We need to include all confounding variables when testing the causal effect of covariates and response variable.

With most covariates in this dataset, you can find regression specifications for which the sign on the estimate will even flip.

As we also saw in Q4, the slope of INDUS on MEDV may decrease conditional on the value of NOX. One way that we may encode for this in a regression model is via **interaction** functions of INDUS and NOX.

Q7: Run and summarize a regression of INDUS, NOX, and INDUS*NOX on MEDV.

```
## YOUR CODE HERE ###
model3 = lm(medv ~ indus + nox + indus:nox, data= boston_housing)
summary(model3)

##
## Call:
## lm(formula = medv ~ indus + nox + indus:nox, data = boston_housing)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -12.45  -4.92  -1.63   3.08  32.24
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  31.438     5.060     6.21 1.1e-09 ***
## indus        -0.293     0.355    -0.83  0.41
## nox         -5.251    10.502    -0.50  0.62
## indus:nox     -0.402     0.651    -0.62  0.54
```

```
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 8.03 on 502 degrees of freedom
## Multiple R-squared:  0.243, Adjusted R-squared:  0.238
## F-statistic: 53.6 on 3 and 502 DF,  p-value: <2e-16
```

Indeed, you will find a negative coefficient on the interaction term (though all terms are now insignificant), indicating that the slope of `INDUS` is decreasing by approximately -0.04 for each increase in `NOX` of 0.1. We will explore interactions in more detail later in the course.

You finished!